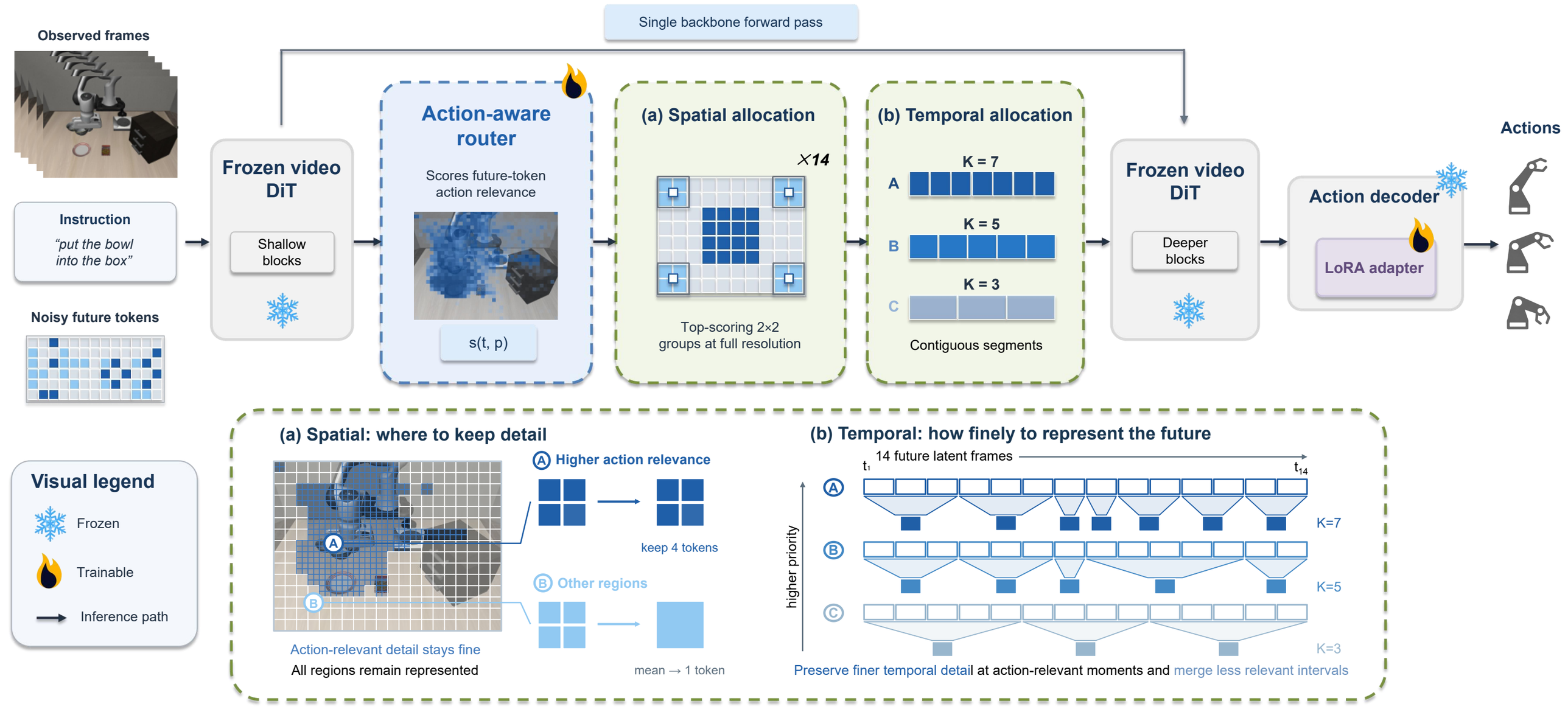

Observed frames
Instruction
"put the bowl into the box"
Noisy future tokens
Single backbone forward pass
Frozen video DiT
Shallow blocks
Action-aware router
Scores future-token action relevance
s(t, p)
(a) Spatial allocation
×14
Top-scoring 2×2 groups at full resolution
(b) Temporal allocation
A
K = 7
B
K = 5
C
K = 3
Contiguous segments
Frozen video DiT
Deeper blocks
Action decoder
LoRA adapter
Actions
Visual legend
Frozen
Trainable
Inference path
(a) Spatial: where to keep detail
A Higher action relevance
keep 4 tokens
B Other regions
mean → 1 token
Action-relevant detail stays fine
All regions remain represented
(b) Temporal: how finely to represent the future
14 future latent frames
t1
t14
higher priority
K=7
K=5
K=3
Preserve finer temporal detail at action-relevant moments and merge less relevant intervals